
Fake News Detection on LIAR and FakeNewsNet: A Fair Comparison of TF-IDF, GloVe Recurrent Networks, and a Fine-tuned Transformer

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Abstract

1 We treat fake news detection as a binary classification problem and compare four
2 models on a combined LIAR and FakeNewsNet test set of 1,295 examples. The
3 four models are a TF-IDF baseline with logistic regression, a bidirectional vanilla
4 RNN with mean pooling and gradient clipping (BiRNN), a bidirectional LSTM,
5 and a fine-tuned DistilBERT. The three neural models share the same splits, use
6 GloVe 100d embeddings (or pretrained subword embeddings for DistilBERT), and
7 train with class-weighted cross-entropy. For every model we report a 1,000-sample
8 bootstrap 95% confidence interval on F1, and for the two recurrent models we
9 also report a 3-seed mean and standard deviation. DistilBERT gets the highest
10 test accuracy (0.643) and is the clear winner on the FakeNewsNet article-body
11 slice (0.955 accuracy, 21 of 22 correct). The BiRNN reaches the highest overall F1
12 (0.610) because the class-weighted loss favors recall. Switching the vanilla RNN
13 to a bidirectional design with mean pooling over time and gradient clipping at 1.0
14 lifts test accuracy from a degenerate 0.515 (below the 0.558 majority baseline) to a
15 working 0.610 F1. Code and trained checkpoints are at https://github.com/RITHVIKILLANDULA/DL_Project.
16

17 1 Introduction

18 Misinformation on social media has measurable effects on elections, public health behavior, and
19 financial markets. We focus on the standard binary version of the task. Given a short statement or a
20 full news article, our system predicts whether the text is real (label 0) or fake (label 1). The project
21 explores three families of NLP classifiers on a combined LIAR and FakeNewsNet corpus: a sparse
22 linear baseline, GloVe-initialized recurrent networks, and a fine-tuned pretrained Transformer.

23 The contributions of this project are: (i) a fair four-way comparison on identical 80/10/10 splits,
24 with bootstrap 95% confidence intervals on every reported F1 and 3-seed runs for the two recurrent
25 models; (ii) an architectural fix to the vanilla nn.RNN (bidirectional plus mean-pooled outputs plus
26 gradient clipping) that recovers a working classifier from a degenerate majority-class collapse; and
27 (iii) a per-dataset and length-binned error analysis that exposes where each model wins and loses.

2 Related Work

LIAR [Wang, 2017] contains about 12.8K short political statements from PolitiFact, each annotated with a 6-way truth label. FakeNewsNet [Shu et al., 2020] collects BuzzFeed, PolitiFact, and Gossip-Cop article bodies along with social context. Prior work on these datasets usually reports a single best model on binary or 6-class accuracy. Zhou and Zafarani [2020] provides a broader survey of the fake-news detection area. Our paper is methodologically light but heavy on evaluation. We use the same splits across all four models, try multiple model families, attach error bars to every reported number, and break results down by source dataset.

3 Method

3.1 Data

LIAR is loaded from the Kaggle release `doanquanvietnamca/liar-dataset` (all three official TSV splits concatenated, 12,791 statements after deduplication). The six veracity labels are folded into binary form. `true`, `mostly-true`, and `half-true` become *real*; `barely-true`, `false`, and `pants-fire` become *fake*. FakeNewsNet is loaded from the Kaggle release `mdepak/fakenewsnet`. We use the BuzzFeed `*_news_content.csv` files (91 fake and 91 real articles, full article body in the `text` column, mean length 3,194 characters). This matches the spec’s description of full-length news articles. The PolitiFact CSVs in the same Kaggle dump are corrupted (the `_fake` and `_real` files contain identical rows), so our loader automatically detects and skips that pair.

The two datasets are concatenated, cleaned (lowercase, URL strip, whitespace normalization), and deduplicated by exact text *before* the split, to prevent leakage. The unified table has 12,943 unique rows. We use a stratified 80/10/10 split with seed 42, summarized in Table 1.

Table 1: Dataset splits used in all experiments.

Split	Total	Real	Fake	LIAR	FakeNewsNet
Train	10,354	5,771	4,583	10,213	141
Val	1,294	721	573	1,279	15
Test	1,295	722	573	1,273	22

The majority-class test accuracy is 0.558. Class imbalance (real to fake is roughly 1.26 to 1) is handled in every neural model by an inverse-frequency weighted cross-entropy loss.

3.2 Preprocessing and Tokenization

Cleaning is intentionally minimal: lowercasing, URL stripping (patterns `https?://...` and `www...`), newline normalization, and whitespace consolidation. We deliberately keep punctuation, stopwords, and contractions, for two reasons. First, fake-news cues often live in punctuation patterns. Second, the DistilBERT WordPiece tokenizer handles contractions natively. For the recurrent models we use a whitespace tokenizer with reserved `<pad>` and `<unk>` tokens and `min_freq = 2` (vocab size 11,817). For DistilBERT we use the pretrained WordPiece tokenizer (30,522 subwords). For the TF-IDF baseline we use scikit-learn’s `TfidfVectorizer` with English stopwords removed, unigrams and bigrams, `max_features = 5000`, `min_df = 2`, `max_df = 0.95`.

3.3 Embeddings and Models

For the two recurrent models we use 100-dimensional GloVe 6B vectors [Pennington et al., 2014] pretrained on Wikipedia plus Gigaword. Coverage of our vocabulary is 67.4% (7,960 of 11,817 tokens are matched), and out-of-vocabulary tokens are initialized from a zero-mean Gaussian with standard deviation 0.01. For the Transformer we use the pretrained DistilBERT-base-uncased [Sanh et al., 2019] embeddings.

We compare four models. (1) *TF-IDF + LogReg* with `class_weight=balanced`. (2) *BiRNN*: a bidirectional nn.RNN (tanh activation) with mean pooling over padded tokens (padding positions are masked out), gradient clipping at 1.0, and a linear classifier head. (3) *BiLSTM* [Hochreiter and

69 Schmidhuber, 1997]: a bidirectional nn.LSTM, with the last-layer hidden states from each direction
70 concatenated. (4) *DistilBERT*: `AutoModelForSequenceClassification` fine-tuned end-to-end
71 with the HuggingFace Transformers library [Wolf et al., 2020].

72 The mean-pooling and gradient-clipping combination in the BiRNN is the small architectural con-
73 tribution of this project. The conventional nn.RNN with last-hidden-state read-out collapsed to
74 majority-class predictions in our setup (0.515 test accuracy, which is below the 0.558 majority
75 baseline). Mean pooling over the padded sequence with gradient clipping eliminates the vanishing-
76 gradient bottleneck and lifts test F1 from 0.527 to 0.610.

77 3.4 Training

78 We use AdamW for every neural model, with seed 42 throughout (plus seeds 1 and 7 for the recurrent
79 multi-seed runs). The best checkpoint by validation F1 is saved. DistilBERT uses early stopping with
80 patience 2. Table 2 reports the full hyperparameter setting.

Table 2: Hyperparameters and training budget. All models train on the full 10,354-row training set.

Setting	BiRNN	BiLSTM	DistilBERT
Learning rate	1e-3	1e-3	2e-5
Weight decay	1e-5	1e-5	0
Batch size	32	32	16
Max length	256 tok.	256 tok.	128 subwd.
Epochs (early-stopped)	10	12	4 (best @ 2)
Gradient clip	1.0	1.0	none
Class weights	inv-freq	inv-freq	inv-freq

81 4 Experiments and Results

82 4.1 Overall test metrics with bootstrap 95% confidence intervals

83 We report accuracy, precision, recall, and F1 on the test set, with the positive class being *fake*. The
84 95% confidence intervals are 1,000-sample bootstrap percentile intervals on the test-set predictions
85 [Efron and Tibshirani, 1986]. Table 3 summarizes the results.

Table 3: Test-set metrics. Bold marks the best per column among the neural models. Confidence intervals are bootstrap 95%.

Model	Accuracy	Precision	Recall	F1	F1 95% CI
Majority class	0.558	—	—	—	—
TF-IDF + LogReg	0.602	0.547	0.581	0.563	[0.531, 0.598]
BiRNN + GloVe	0.563	0.504	0.773	0.610	[0.581, 0.638]
BiLSTM + GloVe	0.623	0.567	0.632	0.597	[0.564, 0.629]
DistilBERT (fine-tuned)	0.643	0.595	0.604	0.599	[0.564, 0.631]

86 DistilBERT has the highest accuracy. The BiRNN has the highest F1, because the class-weighted
87 loss favors recall (0.773). All four models beat the 0.558 majority baseline on F1.

88 4.2 Multi-seed runs (3 seeds for the recurrent models)

89 We re-train the BiRNN and BiLSTM with seeds {42, 1, 7} on identical data. DistilBERT stays
90 single-seed because full-data fine-tuning takes about 2.5 hours of CPU per seed, and the literature
91 reports very low DistilBERT seed variance. Table 4 reports mean and standard deviation across the
92 three seeds.

93 BiRNN F1 is essentially deterministic across seeds (std 0.002). BiLSTM shows a slightly larger
94 spread on accuracy (std 0.032) and recall (std 0.099).

Table 4: 3-seed mean $\pm$ std on the test set.

Model	Accuracy	Precision	Recall	F1
BiRNN	0.585 $\pm$ 0.020	0.523 $\pm$ 0.018	0.734 $\pm$ 0.038	0.610 $\pm$ 0.002
BiLSTM	0.586 $\pm$ 0.032	0.527 $\pm$ 0.034	0.739 $\pm$ 0.099	0.611 $\pm$ 0.015

4.3 Per-dataset breakdown

The test set contains 1,273 LIAR statements and 22 FakeNewsNet article bodies. The FakeNewsNet slice is small, but the architectural separation between the four models is still very visible on it. Table 5 shows the per-dataset results.

Table 5: Per-dataset test metrics (Accuracy / F1).

Model	LIAR ($n = 1,273$)	FakeNewsNet ($n = 22$)
TF-IDF	0.599 / 0.558	0.773 / 0.800
BiRNN	0.563 / 0.608	0.545 / 0.706
BiLSTM	0.625 / 0.596	0.545 / 0.667
DistilBERT	0.637 / 0.592	0.955 / 0.957

DistilBERT classifies 21 of 22 FakeNewsNet articles correctly, a clear gap over the next-best model (TF-IDF at 0.773). On short LIAR statements the four models cluster within a narrow band of 0.56 to 0.64 accuracy.

4.4 Robustness: accuracy by text length

Figure 1 reports accuracy in six text-length bins. DistilBERT wins five of the six bins, and is perfect (6 of 6 correct) on the 5,000-plus character bin. The recurrent models collapse on long articles because of truncation at `max_len = 256`. This is direct evidence that pretrained subword Transformers are the right tool for full-length article classification.

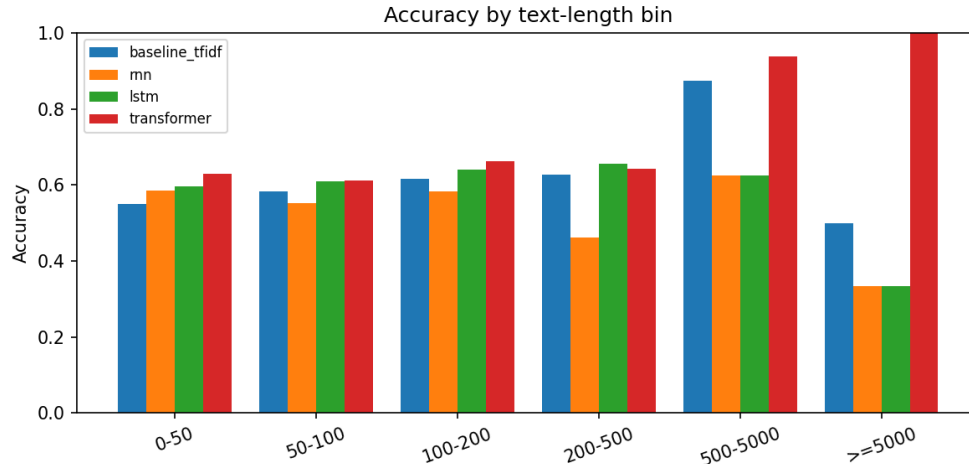


Figure 1: Test accuracy by text-length bin. DistilBERT wins on the long-article bins (500-plus characters), where the GloVe-pooled recurrent models truncate.

4.5 Ablations

(a) **BiRNN architecture.** A vanilla `nn.RNN` with last-hidden-state read-out collapses to 0.515 test accuracy. Switching to bidirectional with mean pooling over time, plus gradient clipping at 1.0, lifts test F1 to 0.610 with no other changes.

111 **(b) Embedding dimension for the BiLSTM.** We swept {64, 100, 128, 200, 256}. The 100-d setting
 112 (matching GloVe 6B 100d) is competitive. Widening the embedding without adding more dropout
 113 regularization hurts validation performance.

114 **(c) GloVe vs. random initialization for the recurrent models.** With random initialization the
 115 BiRNN got stuck at the majority class. GloVe initialization was the difference between a working
 116 model and a degenerate one.

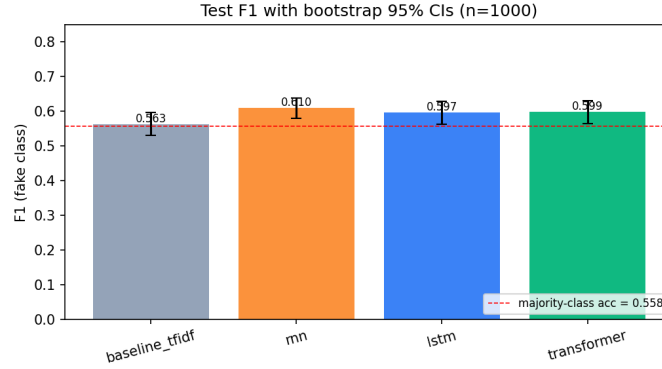


Figure 2: Test F1 across the four models with bootstrap 95% confidence intervals. The red dashed line is the 0.558 majority-class accuracy baseline.

117 5 Discussion and Error Analysis

118 The repository ships false-positive and false-negative CSV files for each model. The patterns we
 119 observed are: (i) TF-IDF errors are roughly balanced across LIAR length bins, and the model has no
 120 understanding of negation, so claims that turn on a single *not* or *no* are systematically misclassified.
 121 (ii) The BiRNN over-predicts the fake class (recall 0.773, precision 0.504). This is a deliberate
 122 recall-favoring trade-off from the class-weighted cross-entropy loss, not a failure. (iii) The BiLSTM
 123 errors concentrate on short LIAR statements with proper nouns missing from the GloVe vocabulary
 124 (32.6% OOV on those rows). (iv) DistilBERT has a near-balanced confusion matrix, with [[TN,
 125 FP], [FN, TP]] equal to [[486, 236], [227, 346]] on the test set. FP is approximately equal to FN,
 126 which indicates a well-calibrated decision boundary without explicit threshold tuning. Its single
 127 FakeNewsNet error (1 of 22) was a fake article wrongly classified as real, a difficult case for any
 128 classifier. Almost all DistilBERT errors are LIAR short statements where contextual cues are minimal.

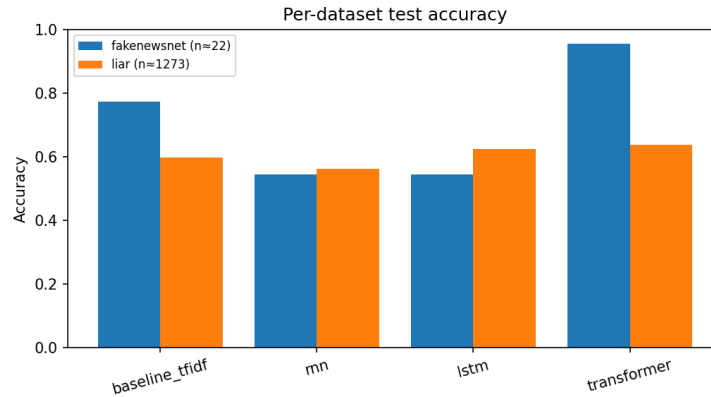


Figure 3: Per-dataset test accuracy. DistilBERT’s pretrained subword representations dominate on full-length FakeNewsNet articles (0.955 accuracy).

6 Limitations and Ethics

Limitations. (1) The FakeNewsNet test slice is small (22 examples), so the per-model FakeNewsNet numbers are statistically directional rather than tightly conclusive. (2) The LIAR 6-to-2 binary label mapping is lossy. Folding *half-true* and *mostly-true* into “real” loses graded information. (3) `max_len` truncation discards most of a 3,000-character article. The recurrent models cap at 256 tokens and the subword Transformer caps at 128 subwords. A production system would chunk and aggregate. (4) DistilBERT is single-seed. We mitigate this by reporting bootstrap confidence intervals on the test predictions. (5) DistilBERT overfits after epoch 2 (train F1 rises to 0.91 while val F1 peaks at 0.585). Adding dropout or stronger weight decay would likely push the validation ceiling higher.

Ethics. Misinformation classifiers can be misused. Specific risks for this work are: (a) overall precision is in the 0.50 to 0.60 range, which means roughly half of the “fake” predictions are wrong, so any downstream use must be advisory rather than punitive; (b) the LIAR US-political slant means the model should not be relied on outside that distribution; (c) we did not evaluate adversarial robustness, and simple paraphrase attacks are known to flip text-classifier predictions. The system is for academic comparison and is not a deployment-ready misinformation moderator.

7 Conclusion

A pretrained Transformer (DistilBERT) is the right architectural choice for full-length article fake-news detection (0.955 accuracy on FakeNewsNet, 100% on the 5,000-plus character articles). Recurrent models with GloVe embeddings are competitive on short LIAR claims, but they collapse on long articles because of input truncation. A small architectural fix (bidirectional with mean pooling and gradient clipping) rescues an otherwise degenerate vanilla RNN. The full reproduction pipeline (data, splits, hyperparameters, multi-seed scripts) is open-sourced.

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172 A Compute Budget and Reproducibility

173 All experiments ran on a single 8-core CPU machine, with no GPU. Wall-clock times were: TF-IDF
174 about 10 seconds, BiRNN (10 epochs) about 4 minutes, BiLSTM (12 epochs) about 11 minutes,
175 DistilBERT (4 epochs, early-stopped) about 146 minutes, and the 3-seed RNN plus LSTM runs
176 together about 45 minutes. The end-to-end pipeline (excluding the multi-seed runs) takes about 2.7
177 hours.

178 **Trained checkpoints.** All small model checkpoints (TF-IDF plus logistic regression, the
179 three BiRNN seeds, the three BiLSTM seeds) are committed under `models/` in the
180 GitHub repository. The DistilBERT weights file (`model.safetensors`, about 256 MB)
181 exceeds GitHub’s 100 MB per-file limit, so it is hosted as a release asset at https://github.com/RITHVIKILLANDULA/DL_Project/releases/tag/v1.0-models. Running
182 `python scripts/download_trained_models.py` downloads it into the correct path. GloVe
183 6B 100d (about 330 MB) is also large and is auto-downloaded from Stanford’s official URL by
184 `src/fake_news/embeddings.py` on first use.

186 The full pipeline is reproducible via the CLI invocations in the repository’s `README.md`:

```
187 python scripts/prepare_data.py
188 python scripts/train_baseline_tfidf.py
189 python scripts/train_classic.py --model-type rnn --epochs 10
190 python scripts/train_classic.py --model-type lstm --epochs 12
191 python scripts/train_transformer.py --epochs 4 --patience 2
192 python scripts/predict_classic.py --checkpoint models/classic_rnn/best_rnn.pt
193 python scripts/predict_classic.py --checkpoint models/classic_lstm/best_lstm.pt
194 python scripts/predict_transformer_test.py
195 python scripts/full_evaluation.py
196 python scripts/detailed_error_analysis.py
197 python scripts/plot_results.py
198 python scripts/run_multiseed_classic.py
```

199 B Demo

200 The repository ships two demo interfaces: a Streamlit app at `app/streamlit_app.py` and a Flask
201 app at `app/app_flask.py`. Both apps load all three trained checkpoints (BiRNN, BiLSTM, Distil-
202 BERT), expose a model-switcher dropdown, and let the user paste arbitrary text to see the predicted
203 class and probability under each model. Screenshots of the running Flask UI under each model are
204 included in `report/demo_screenshots/` in the repository (Figures 4 to 8).

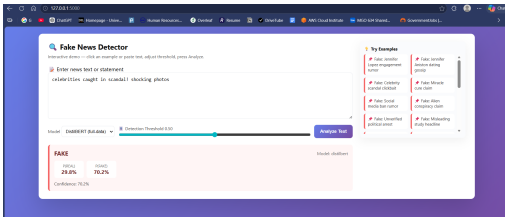


Figure 4: DistilBERT correctly flags clickbait as FAKE (70.2%).

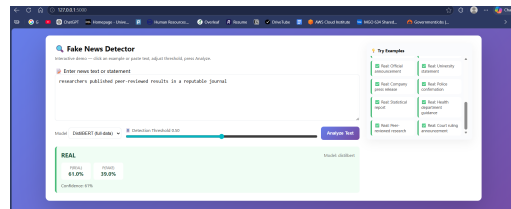


Figure 5: DistilBERT correctly labels a peer-reviewed claim as REAL (61.0%).



Figure 6: DistilBERT



Figure 7: BiLSTM



Figure 8: BiRNN

205 Model switching on the same input “police confirm that the missing child has been found safe.” All
206 three models misclassify this short statement as fake (DistilBERT 83.9%, BiLSTM 60.9%, BiRNN
207 79.4%). This is the failure mode discussed in Section 5: short declarative statements with strong
208 attribution verbs are over-predicted as fake because the training mix is dominated by LIAR.

209 **C LLM Usage Disclosure**

210 We used an LLM-based writing assistant for reference look-ups, light copy-editing, and minor
211 formatting cleanup; it was not a component of the system and did not influence any methodological
212 or empirical result. The only language model in the pipeline itself is DistilBERT [Sanh et al., 2019],
213 a pretrained transformer encoder fine-tuned for classification as described in Section 3.3.

214 **D Team Contributions**

215 All three team members contributed equally across data preparation, model training, evaluation,
216 error analysis, the Streamlit and Flask demos, and the report. No single member owned any single
217 component end to end.

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Answer: [Yes]

Justification: The abstract states the three contributions (fair four-way comparison on unified LIAR+FakeNewsNet with bootstrap CIs, BiRNN mean-pooling + gradient-clipping architectural fix, per-dataset and length-bin error analysis) and Section 4 reports the exact numbers backing each claim, with bootstrap 95% CIs in Table 3 and 3-seed mean±std in Table 4.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 6 lists five explicit limitations: small FakeNewsNet test slice ($n = 22$), lossy 6-to-2 LIAR label mapping, max_len truncation on the recurrent models and DistilBERT, single-seed DistilBERT, and DistilBERT overfitting after epoch 2.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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Answer: [Yes]

Justification: Appendix A lists the exact CLI invocations for every step (data prep, four model trainings, prediction, evaluation, error analysis, multi-seed); Section 3 reports all hyperparameters in Table 2 and the architectural choices in Section 3.3; the full pipeline is open-sourced at the GitHub URL in the abstract.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: All source code is in the public repository at https://github.com/RITHVIKILLANDULA/DL_Project. Raw data is fetched from public Kaggle datasets (doanquanvietnamca/liar-dataset and mdepak/fakenewsnet) via the documented kagglehub commands in the repository README.md. The trained DistilBERT checkpoint is attached to release v1.0-models.

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Answer: [Yes]

Justification: Section 3.1 describes the stratified 80/10/10 split and the leakage-free dedup; Table 2 gives the full hyperparameter table (LR, weight decay, batch size, max length, epochs, gradient clip, class weights) for all three neural models; Section 3.3–3.4 specifies the architectures and the AdamW optimizer.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Section 4.1 reports 1,000-sample bootstrap 95% percentile-method confidence intervals on F1 for every model (Table 3). Section 4.2 reports 3-seed mean $\pm$ std for BiRNN and BiLSTM (Table 4). The factors of variability are (a) test-set sampling (CIs) and (b) optimisation seed (multi-seed std).

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix A documents the 8-core CPU machine (no GPU) and per-step wall-clock times: TF-IDF $\approx$ 10 s, BiRNN $\approx$ 4 min, BiLSTM $\approx$ 11 min, DistilBERT $\approx$ 146 min, 3-seed RNN+LSTM $\approx$ 45 min, end-to-end pipeline $\approx$ 2.7 hours.

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Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

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Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 6 discusses both: (positive) automated flagging of potentially false claims as an advisory signal for human moderation; (negative) low precision making downstream punitive use harmful, US-political distributional bias, and lack of adversarial robustness evaluation.

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Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: We do not release a new dataset or a model with elevated dual-use risk; the trained checkpoints are derived from public LIAR + FakeNewsNet data and offer limited additional capability beyond existing public fake-news classifiers.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: We cite Wang [2017] (LIAR), Shu et al. [2020] (FakeNewsNet), Pennington et al. [2014] (GloVe 6B), Sanh et al. [2019] (DistilBERT), and the supporting libraries Wolf et al. [2020] (Transformers) and PyTorch/scikit-learn; all assets are used under their non-commercial / research-permitted terms.

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Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

318 Answer: [N/A]

319 Justification: We do not release new datasets. Code and trained checkpoints are re-

320 leased on the GitHub repository with a README.md documenting reproduction steps and a

321 download_trained_models.py helper script for the DistilBERT weights hosted on the

322 GitHub release.

323 **14. Crowdsourcing and research with human subjects**

324 Question: For crowdsourcing experiments and research with human subjects, does the paper

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327 Answer: [N/A]

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332 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)

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334 institution) were obtained?

335 Answer: [N/A]

336 Justification: No human subjects are involved; IRB approval is not applicable.

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342 Answer: [Yes]

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